

CLAIRE SHEN

Active DoD Secret Clearance

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PROFESSIONAL SUMMARY

Artificial Intelligence student at UC San Diego integrating computer vision and agentic workflow in DoD test and evaluation environments, with an Active Secret Security Clearance. Experience spans deep learning, edge deployment, and dataset creation across defense and healthcare applications.

EDUCATION

University of California, San Diego – Jacobs School of Engineering

San Diego, CA

B.S. Artificial Intelligence

June 2029

Relevant Coursework: Intro to AI, Data Structures, Object-Oriented Programming, Linear Algebra

EXPERIENCE

AI/ML Engineering Intern (NREIP)

Norco, CA

Naval Surface Warfare Center (NSWC) Corona | Range Systems Engineering

June 2026 – August 2026

- Built local video annotation tool for frame-by-frame data labeling, including automated labeling with a human-in-the-loop pipeline, file self-management, and dataset review and export features
- Designed PDF ingestion pipeline using Docling to predict malfunctions from corpus of unstandardized part-failure reports
- Optimized YOLO model running on real-time video, increasing throughput 4x and mAP by ~4%

Engineering Intern (SEAP)

Fallbrook, CA

NSWC Corona Fallbrook Detachment | Rapid Prototyping Lab

June 2025 – August 2025

- Developed AI-enabled vehicle damage assessment app for Marines with a team of 4 interns
- Collected, cleaned, and labeled ~1,000 images of vehicle corrosion to build a custom dataset
- Built and tested deep learning models for image segmentation on Raspberry Pi and Google Edge TPU

Engineering Intern (SEAP)

Fallbrook, CA

NSWC Corona Fallbrook Detachment | Rapid Prototyping Lab

June 2024 – August 2024

- Designed, built, and FEA tested low-cost mobile tools using additive manufacturing to support Marines
- Reverse-engineered vehicle parts in Fusion 360 and SOLIDWORKS for metal 3D printing

PROJECTS

DermLens.ai | Automated Monitoring for Skin Cancer Treatment

September 2023 – Present

- Build and evaluate CNNs on 8,000+ clinical images of skin cancer treatment
- Train machine learning models to grade redness and track patient progress during treatment

ACTIVITIES & LEADERSHIP

Affiliations: Women in Defense (WID); Society of Women Engineers (SWE); Women in Computing (WiC)

SWENext Club of Murrieta Valley High School — President & Founder

August 2023 – June 2025

- Founded SWE chapter; grew club to 40+ students with monthly speakers and mentorship pipeline

MVHS VEX Robotics Team – Co-Captain

August 2022 – June 2025

SKILLS

Languages & Tools: Python, Docker, Git, C/C++, Java, JavaScript, C#

ML Frameworks: PyTorch, TensorFlow, OpenCV, Pandas, Scikit-Learn

Machine Learning: Agentic AI, LLM Orchestration, RAG, Local LLM Inference, Quantization, Computer Vision, Deep Learning, CNNs, Image Segmentation, Object Detection, Model Training & Evaluation, Dataset Creation & Annotation

Embedded ML: Edge Model Deployment, Google Edge TPU, Raspberry Pi

Engineering: SOLIDWORKS, Fusion 360, FEA, Reverse Engineering, Additive Manufacturing, 3D Scanning